

Ontology Development Using Protégé Ontology Editor

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Recap: Ontology

- ▶ The study of things by explaining relationship of concepts within things.

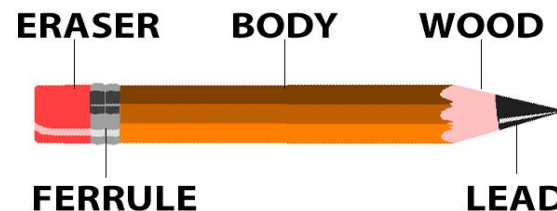


- ▶ What is a 'Pencil'?

- ▶ We can explain it by

- ▶ Defining it : a **writing utensil** with a **graphite lead** embedded in a **wooden shaft**. It is mainly used for writing and artistic drawing.

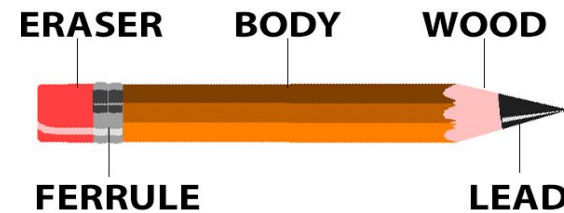
- ▶ Defining parts of it :



Components in an Ontology

▶ Classes or Concepts

- ▶ Entities of things related to the focused topic
- ▶ Such as **pencil**, **lead**, **pencil_body**, **ferrule** and **eraser**.



▶ Relations

- ▶ Linking of entities
- ▶ Such as **lead** is a part of **pencil**, **pencil** is a **writing_utensil**

▶ Individuals or Instances

- ▶ an example or single occurrence of the concept

Ontology Relations

► Relations

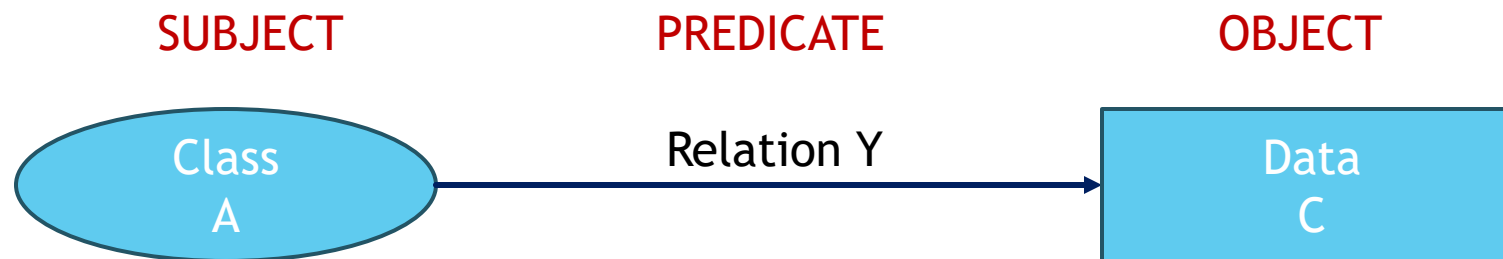
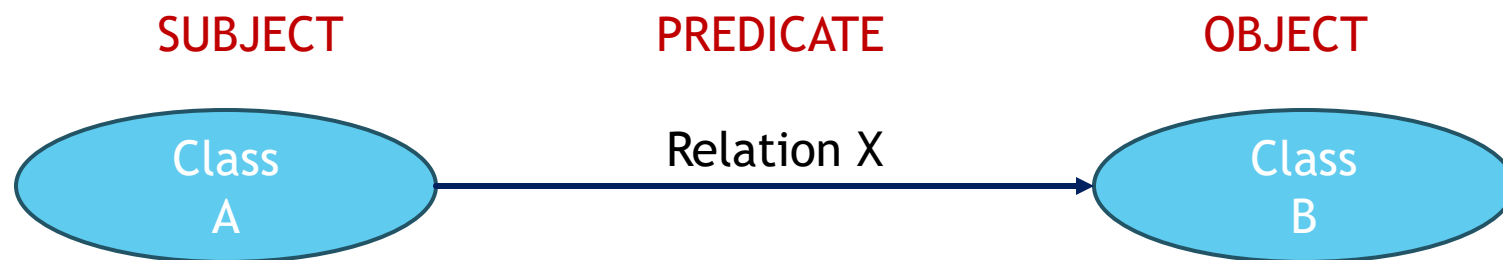
- **Is-A** relation : forming hierarchy tree (superclass-subclass)
 - Acyclic graph
- Property relation : explaining part of the class
 - **Object property (part-of)**: class to class relation
 - Domain: Class
 - Range: Class
 - **Data property (attribute-of)**: class to data relation
 - Domain: Class
 - Range: Data type
- **TypeOf (instance-of)** relation: linking class to individuals



Protégé Ontology Editor Tool

- ▶ Developed by **Center for Biomedical Informatics Research, Stanford University**
 - ▶ free, open source ontology editor and a knowledge management system
 - ▶ supporting for the OWL 2 Web Ontology Language
 - ▶ direct in-memory connections to description logic reasoners including HermiT and Pellet
 - ▶ Current version: 5.5.0+
 - ▶ Download at: <https://protege.stanford.edu/products.php#desktop-protege>

Triple : [SUBJECT]	[PREDICATE]	[OBJECT]
Class	IS-A	Class
Class(Domain)	Object Property	Class(Range)
Class(Domain)	Data Property	Data(Range)
Class	Typeof/Instance of	Individual



Specific Terms

- ▶ **URI & IRI**: Uniformed Resource Identifier and Internationalized Resource Identifier
 - ▶ Classes and relations in ontology are needed to be **specific** and **referable**.
 - ▶ Name of the class is the URI with RDF prefix
- ▶ **RDF**: Resource Description Framework
 - ▶ The marked-up language written in XML to keep ontological details

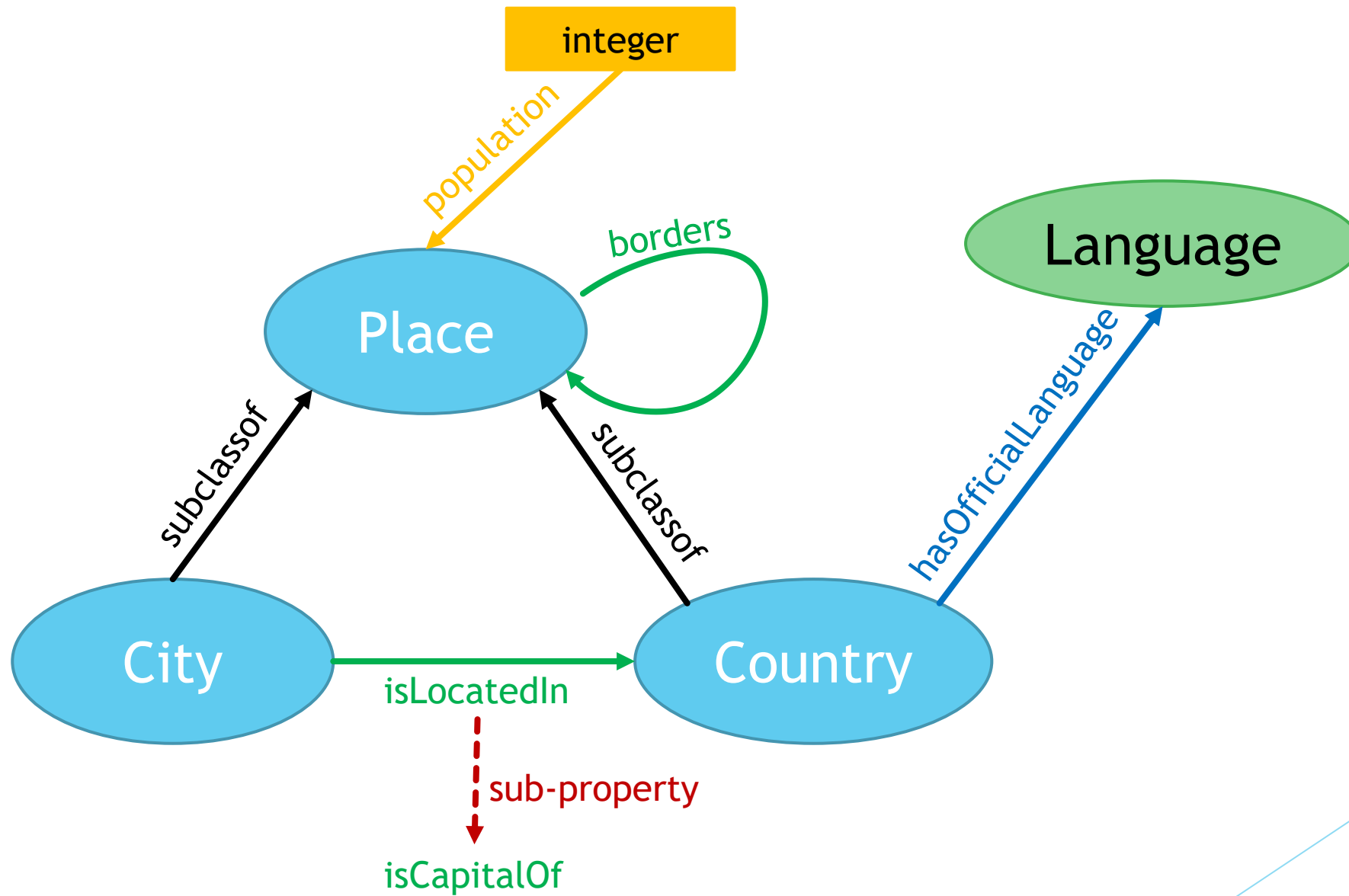
What difference from past classes?

▶ Concept Relationship

- ▶ Not only one concept, but many independent concepts but all linked are in an ontology
 - ▶ Should have many hierarchical trees; tree only represent is-a relation.
- ▶ Part-of and Attribute-of are the type of ontological relation
 - ▶ But they are not the name of the relation, we need to name the relation differently
 - ▶ Role concept → how concept plays a role to another concept
- ▶ Some of conceptualised idea may be out of scope of the ontology objectives

Property Characteristics

Characteristics	Description	example
Functional	A functional property has a single value (object) for a given instance	has_StudentID [John] has_StudentID [64119910]
Inverse Functional	A functional property has a single value (subject) for a given instance If another subject is linked to the object by the predicate, the "other" subject is inferred to actually be the same subject.	has_Child [TonyStark] has_Child [MorganS] [Ironman] has_Child [MorganS] = [TonyStark] isSameAs [Ironman]
Transitive	If a property relates X to Y and Y to Z, then X and Z are also related by that property.	ancestor [John] ancestor [Jim] [Jim] ancestor [Jame] = [John] ancestor [Jame]
Symmetric	If X relates to Y, then Y also relates to X in the same way	border [Thailand] border [Laos] = [Laos] border [Thailand]
Asymmetric	If X relates to Y, then Y is impossible to relate to X in the same way	birth_mother [Jane] birth_mother [Jim]



Country Information

Country	Capital City	Language	Border	Population
Thailand	Bangkok	Thai	Laos, Myanmar, Malaysia, Cambodia	69M
Laos	Vientiane	Lao	Thailand, Myanmar, Vietnam, Cambodia	7.8M
Myanmar	Naypyidaw	Burmese	Thailand, Laos	54M
Malaysia	Kuala Lumpur	Malay	Thailand, Indonesia, Brunei	35M
Indonesia	Jakarta	Indonesian	Malaysia	281M
Vietnam	Hanoi	Vietnamese	Laos, Cambodia	101M
Cambodia	Phnom Penh	Khmer	Thailand, Laos, Vietnam	17M
Brunei	Bandar Seri Begawan	Malay	Malaysia	0.5M
Singapore	Singapore	English	-	6M
Philippines	Manila	Filipino	-	117M

City Information

City	Border	Population
Bangkok	-	10.5M
Chiang Mai	Chiang Rai	1.5M
Chiang Rai	Chiang Mai, Bokeo	1.0M
Ranong	Tanintharyi	0.3M
Vientiane	-	0.9M
Luang Prabang	-	0.1M
Bokeo	Chiang Rai	0.2M
Tanintharyi	Ranong	1.4M
Naypyidaw	-	0.9M
Yangon	-	4.1M
Kuala Lumpur	Petaling Jaya	2.0M
Petaling Jaya	Kuala Lumpur, Shah Alam	0.8M
Shah Alam	Petaling Jaya	0.7M
Jakarta	Bekasi, Depok	10.8M
Bekasi	Jakarta, Depok	2.6M

City	Border	Population
Depok	Jakarta, Bekasi	2.1M
Bogor	-	1.1M
Bandung	Cimahi	2.5M
Cimahi	Bandung	0.6M
Surabaya	-	2.9M
Hanoi	-	8.5M
Da Nang	-	1.2M
Huế	-	0.4M
Phnom Penh	-	2.3M
Singapore	-	6.0M
Manila	Quezon City	1.8M
Quezon City	Manila	3.0M
Cebu	-	0.9M

```
<rdf:Description
rdf:about="http://www.recshop.fake/cd/Empire Burlesque">
  <cd:artist>Bob Dylan</cd:artist>
  <cd:country>USA</cd:country>
  <cd:company>Columbia</cd:company>
  <cd:price>10.90</cd:price>
  <cd:year>1985</cd:year>
</rdf:Description>
```

```
<rdf:Description
rdf:about="http://www.recshop.fake/cd/Hide your heart">
  <cd:artist>Bonnie Tyler</cd:artist>
  <cd:country>UK</cd:country>
  <cd:company>CBS Records</cd:company>
  <cd:price>9.90</cd:price>
  <cd:year>1988</cd:year>
</rdf:Description>
```

You are Petshop Owner. This is the products. Make Ontology based on these using Protege

- **Food**

- **Dog Food**

- Royal Canin - Mini Adult Dry Dog Food — Chicken flavor, small kibble, adult, small breed, 2 kg
 - Pedigree - Puppy Complete Nutrition Dry Food — Chicken and vegetable flavor, small kibble, puppy, 1.5 kg
 - Cesar - Classic Loaf Wet Dog Food — Beef flavor, loaf texture, adult, 100 g

- **Dog Treats**

- Purina - Beggin' Strips — Bacon flavor, soft texture, strip-shaped, 170 g
 - Greenies - Original Dental Treats — Chicken flavor, chewy texture, brush-shaped, medium dogs, 340 g

- **Cat Food**

- Royal Canin - Indoor Adult Dry Cat Food — Chicken flavor, small kibble, adult, indoor cats, 2 kg
 - Whiskas - Tuna Wet Cat Food — Tuna flavor, chunks in gravy, adult, 85 g

- **Cat Treats**

- Temptations - Tasty Chicken Cat Treats — Chicken flavor, crunchy outside/soft inside, pillow-shaped, 85 g

- **Bird Food**

- Vitakraft - Menu Vital Bird Seed Mix — Parakeets, millet/canary seed/sunflower seed, 500 g
 - Kaytee - Forti-Diet Pro Health Bird Food — Parakeets, pellet form, small pellets, 1 kg

- **Grooming**

- **Dog Grooming**

- Earthbath - Oatmeal & Aloe Dog Shampoo — Vanilla and almond scent, cleans/deodorizes, all coat types, 472 mL
 - TropiClean - Gentle Coconut Dog Conditioner — Coconut scent, conditions/softens coat, all coat types, 473 mL

- **Cat Grooming**

- Burt's Bees - Hypoallergenic Cat Shampoo — Shea butter, cleans/moisturizes, all coat types, 296 mL

- **Hygiene**

- Arm & Hammer - Clump & Seal Cat Litter — Clumping clay, odor control, urine absorption, 6.35 kg

- Arcadia Reptile - T5 HO UVB Kit 12% — 24 W, 550 mm, fluorescent tube, for desert reptiles, provides UVB

- Exo Terra - Heat-Glo Infrared Heat Lamp — 75 W, infrared bulb, provides heat, for reptiles and amphibians

- Exo Terra - Digital Hygrometer — digital display, measures relative humidity, 10–99% RH

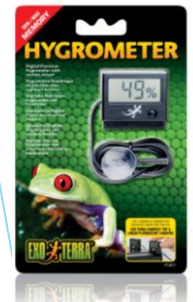
- Zoo Med - Repti Calcium Without D3 — calcium carbonate, powder, phosphorus-free, for reptiles and amphibians, dietary supplement

- Fluval - AquaClear Power Filter — 3-stage filtration, 378 L/h, for freshwater aquariums, removes debris and impurities

- Tetra - Aquarium Air Pump — adjustable airflow, 3.8 L/min, supplies oxygen to aquarium water

- Frontline Plus - Flea & Tick Treatment for Dogs — topical treatment, 3-dose package, for dogs over 8 weeks, controls fleas and ticks

- K9 Advantix II - Flea & Tick Treatment — topical treatment, repels and kills fleas and ticks, for dogs, 4-dose package



Annotation properties

Datatypes

Individuals

Classes

Object properties

Data properties

Class hierarchy: City

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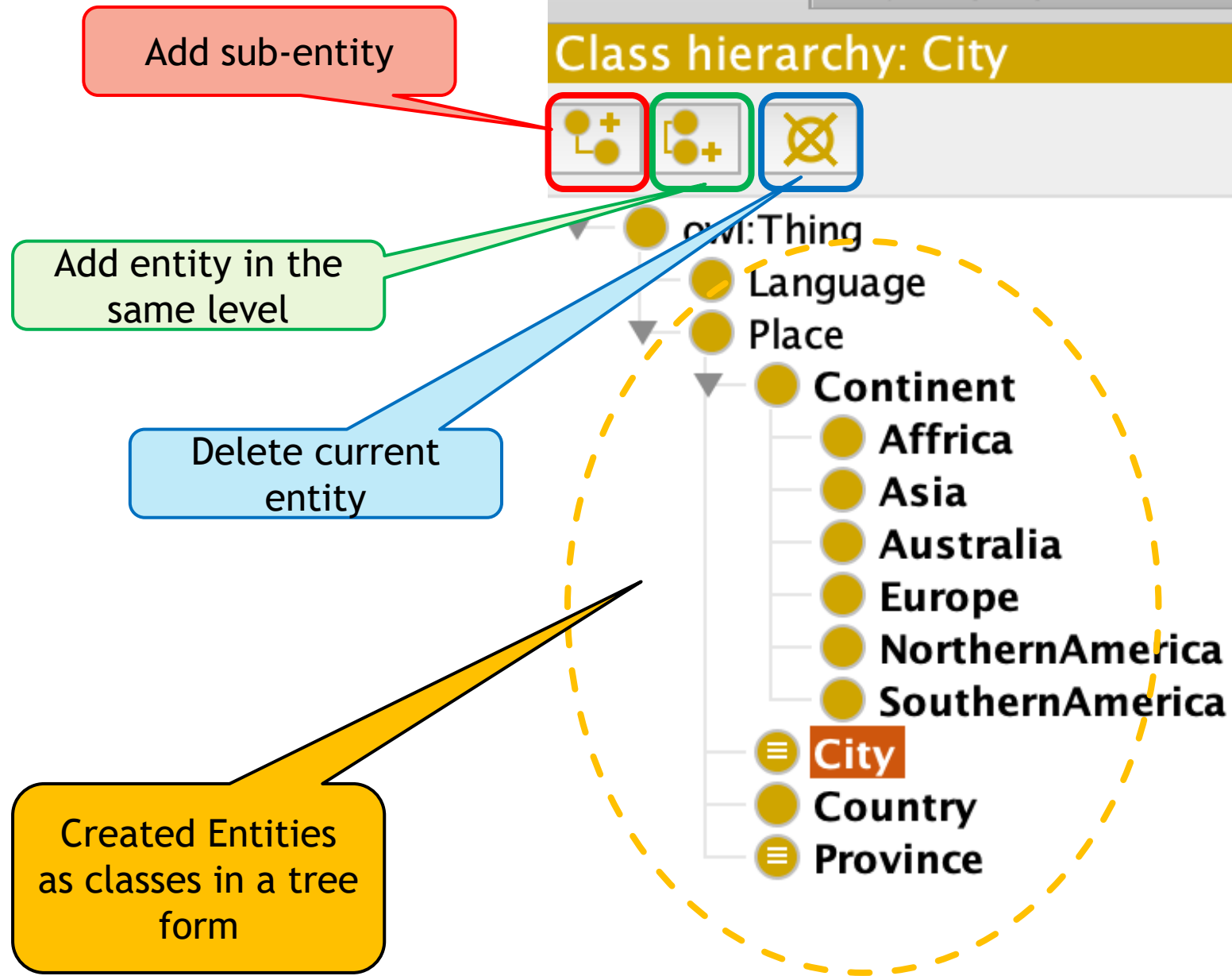
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Asserted

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Add more of this detail

Add annotations: such as comment and label

Delete the current detail

Edit the current detail

Equivalent To

 **City**

SubClass Of

 **Place**

General class axioms

SubClass Of (Anonymous Ancestor)

 **Province**



Ontology header:

Ontology IRI

http://www.semanticweb.org/bonus/ontologies/2019/7/untitled-ontology-

Ontology Version IRI

e.g. http://www.semanticweb.org/bonus/ontologies/2019/7/untitled-ontol

Annotations +

Ontology metrics:

Metrics

Axiom	0
Logical axiom count	0
Declaration axioms count	0
Class count	0
Object property count	0
Data property count	0
Individual count	0
Annotation Property count	0

Class axioms

SubClassOf	0
EquivalentClasses	0
DisjointClasses	0
GCI count	0
Hidden GCI Count	0

Object property axioms

SubObjectPropertyOf	0
ObjectPropertyDomain	0
ObjectPropertyRange	0
ObjectPropertyAxioms	0

Ontology imports | Ontology Prefixes | General class axioms

Imported ontologies:

Direct Imports +

Indirect Imports